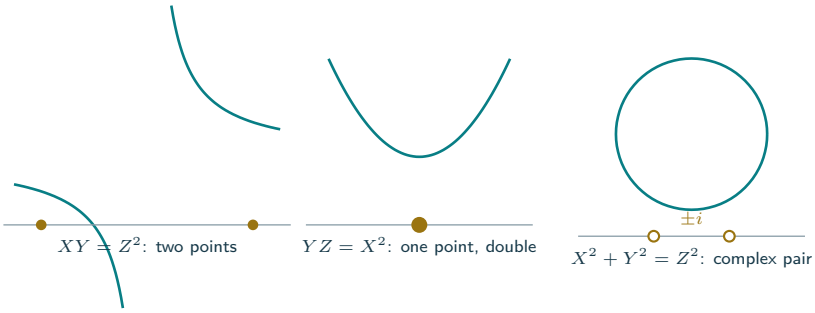
PROJECTIVE GEOMETRY LADDER 3 OF 9

Homogenising curves

Padding terms with powers of Z makes a curve projective and exposes its points at infinity.

1 / THE RULE AND THE PICTURE

A curve of degree d becomes $F = Z^d f(X/Z, Y/Z)$, and $Z = 0$ finds its points at infinity



Replace x by X/Z and y by Y/Z and clear denominators, so that every term has degree d:

$$F(X, Y, Z) = Z^d f\left(\tfrac{X}{Z}, \tfrac{Y}{Z}\right)$$

Setting $Z = 0$ in F finds where the curve meets the line at infinity.

2 / THE STANDARD CURVES

Each curve has its own pattern at infinity

curve	homogeneous equation	points with $Z = 0$
line $y = 2x + 1$	$Y = 2X + Z$	$(1 : 2 : 0)$
parabola $y = x^2$	$YZ = X^2$	$(0 : 1 : 0)$, a double point
hyperbola $xy = 1$	$XY = Z^2$	$(1 : 0 : 0)$ and $(0 : 1 : 0)$
circle $x^2 + y^2 = 1$	$X^2 + Y^2 = Z^2$	$(1 : i : 0)$ and $(1 : -i : 0)$
cubic $y^2 = x^3 - 2$	$Y^2Z = X^3 - 2Z^3$	$(0 : 1 : 0)$, a triple point

3 / ONE CURVE IN THREE DISGUISES

The conics differ only in how they meet the line at infinity

The hyperbola crosses the line at infinity in two points, which are the directions of its asymptotes, the parabola touches it at one point counted twice, and the circle meets it in two complex points $(1 : \pm i : 0)$, the circular points that every circle passes through.
Over the complex numbers a parabola, a hyperbola and a circle are the same curve seen from different positions of the line at infinity, and a projective map can carry one to another.

4 / MULTIPLICITY APPEARS

A root of F at infinity can repeat

For the parabola $Z = 0$ forces $X^2 = 0$, a double root, which records that the line at infinity is tangent to the curve there, and for the cubic $Z = 0$ forces $X^3 = 0$, a triple root, so the point at infinity is a flex.
Repeated roots are how the algebra remembers tangency, and counting them is what makes the intersection counts of the next sheets exact.

5 / TRY THESE

Homogenise and look at infinity

(a) Homogenise $y = x^3$: degree 3 gives $YZ^2 = X^3$, and $Z = 0$ gives $X^3 = 0$, so the point $(0 : 1 : 0)$ with multiplicity 3.
(b) Homogenise $x^2 - y^2 = 1$: $X^2 - Y^2 = Z^2$, with $Z = 0$ giving $(1 : 1 : 0)$ and $(1 : -1 : 0)$, so it is a hyperbola with asymptote slopes ± 1.
(c) Homogenise $x + y = 1$: $X + Y = Z$, a line with the point $(1 : -1 : 0)$ at infinity.

ASIDE / ONE EQUATION, ALL POINTS

The homogeneous equation does not distinguish the plane from the line at infinity, so one polynomial and one count handle all of the curve at once.

LINKS TO OTHER SHEETS

The next sheet counts how many points two such curves share, using exactly these equations.

Convention. d is the degree of f; F is homogeneous of degree d; the circular points are $(1 : \pm i : 0)$.